

Students will perform vulnerability scanning of a given vulnerable machine and export the results

Installing Nessus for networking monitoring:

Download link: <https://www.tenable.com/downloads/nessus?loginAttempted=true>

Selecting correct version:

1 Download and Install Nessus

Choose Download

Version

Nessus - 10.11.1



Platform

Linux - RHEL 8 - x86_64

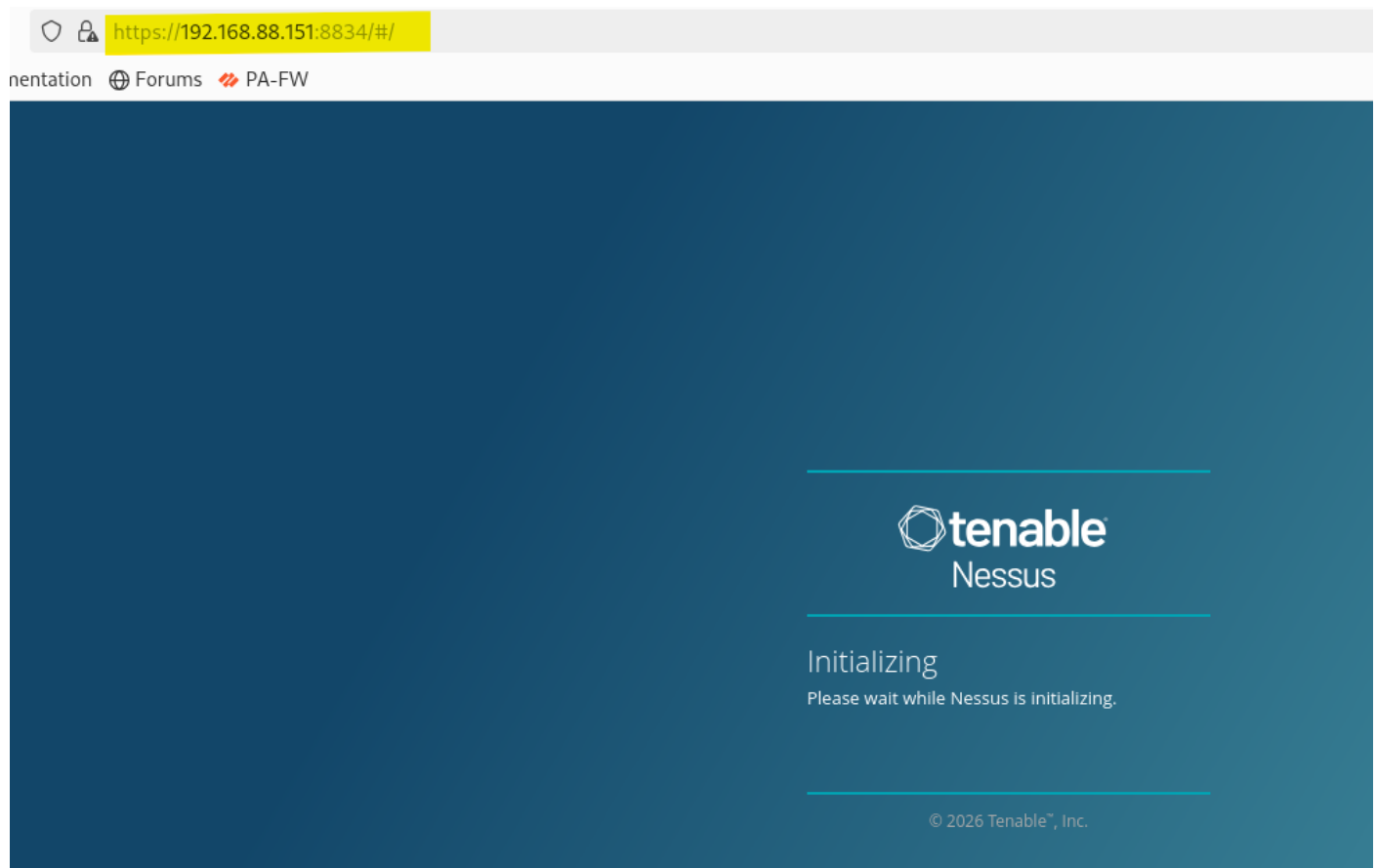


Download

Checksum

- `sudo systemctl start nessusd`
- `sudo systemctl enable nessusd`

accessing Nessus on firefox <https://192.168.88.151:8834/>



Register Nessus Essential for small network



Welcome to Nessus

You can click Settings to configure the Nessus proxy, plugin feed, and encryption password settings before you start the installation, or you can select Register Offline to configure an offline installation.

When you are ready, click Continue to proceed with the installation.

☐ Register Offline

[Settings](#) [Continue](#)

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Get an activation code

To register for a free Nessus Essentials activation code, enter your information.

First Name Last Name

Email

Already have activation code? Skip this step to enter it manually.

[Back](#) [Skip](#) [Register](#)

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Verify Email

Please verify your email address to continue activating your product.

An email was sent to your inbox. Please check your inbox and click the link in the email to verify. Once you have verified your email, you can return to this page and click 'Verify' to continue.

[Back](#) [Verify](#)

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Verify over your email ID and click on verify button.



Email Verification

Your email was verified. You can continue using the application.



License Information

Activation Code: WKZA-DZVE-DLUF-MT57-LBUA

[Continue](#)

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Activation Code: WKZA-DZVE-DLUF-MT57-LBUA



Create a user account

Create a Nessus administrator user account. Use this username and password to log in to Nessus.

Username *

Password *

[Back](#) [Submit](#)



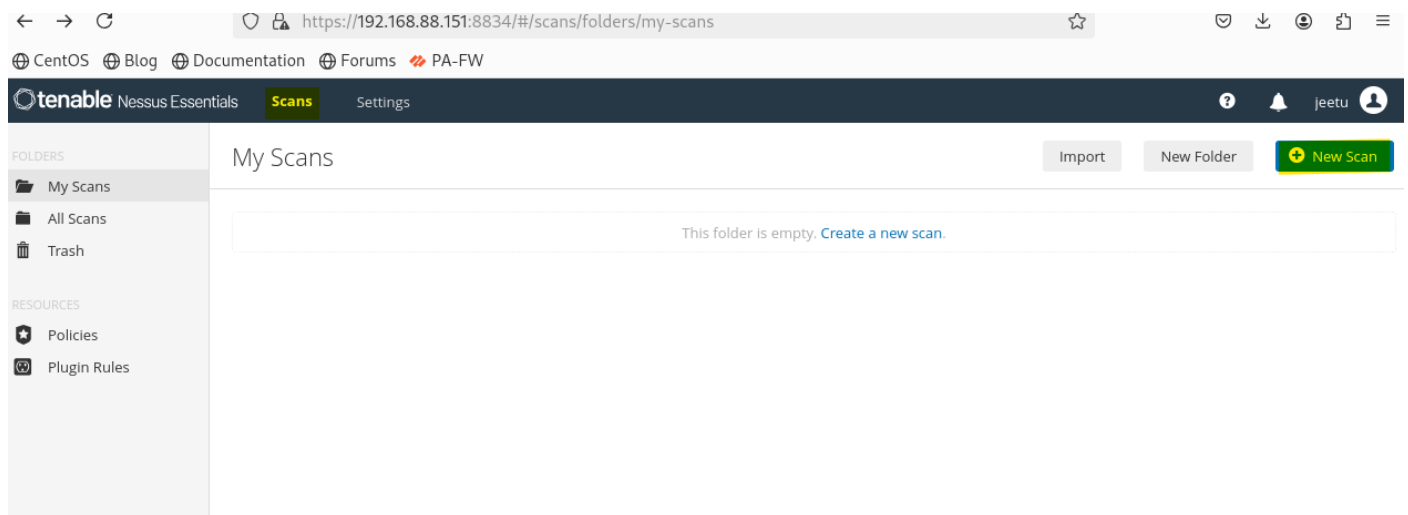
Initializing

Please wait while Nessus is initializing.

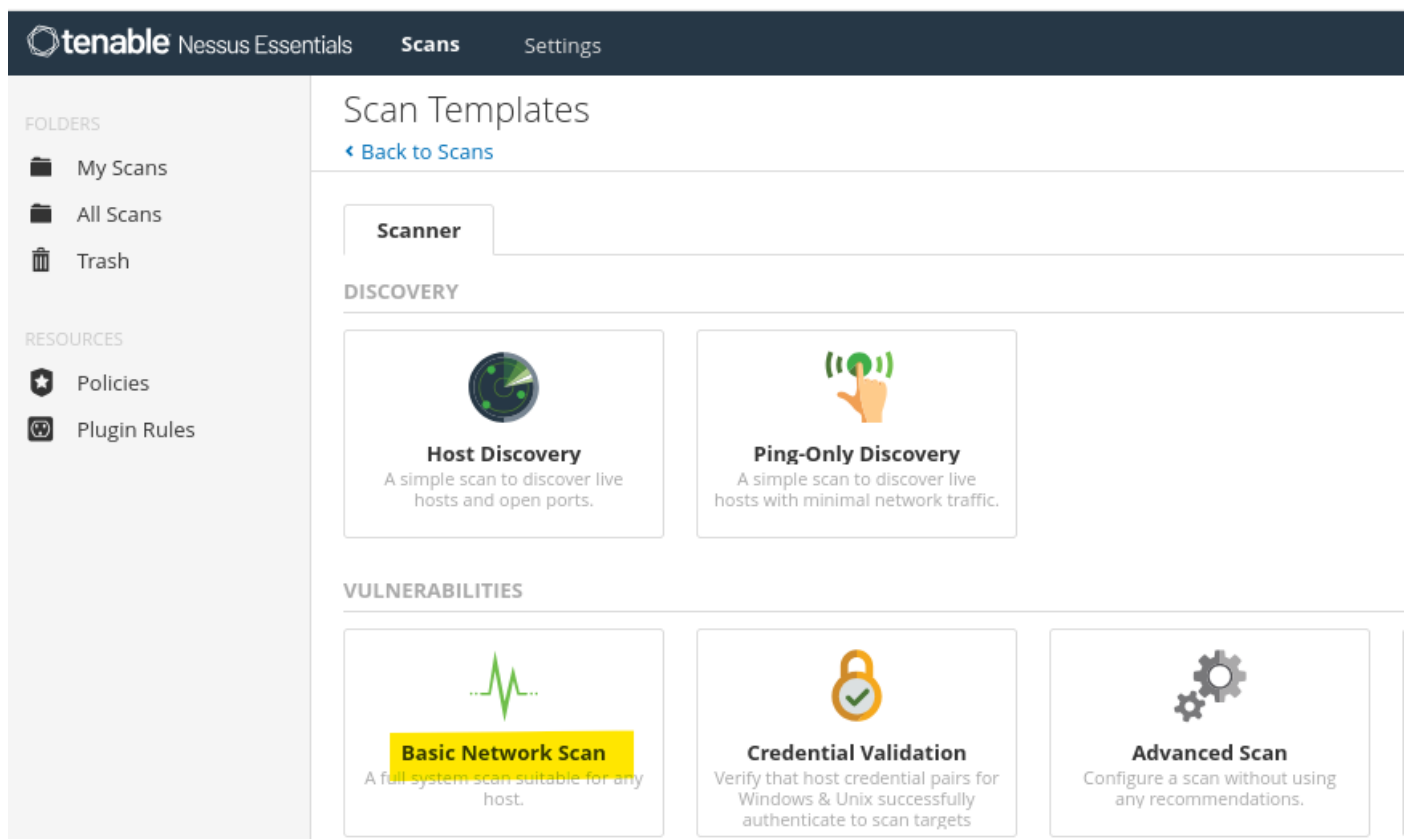
Downloading plugins...

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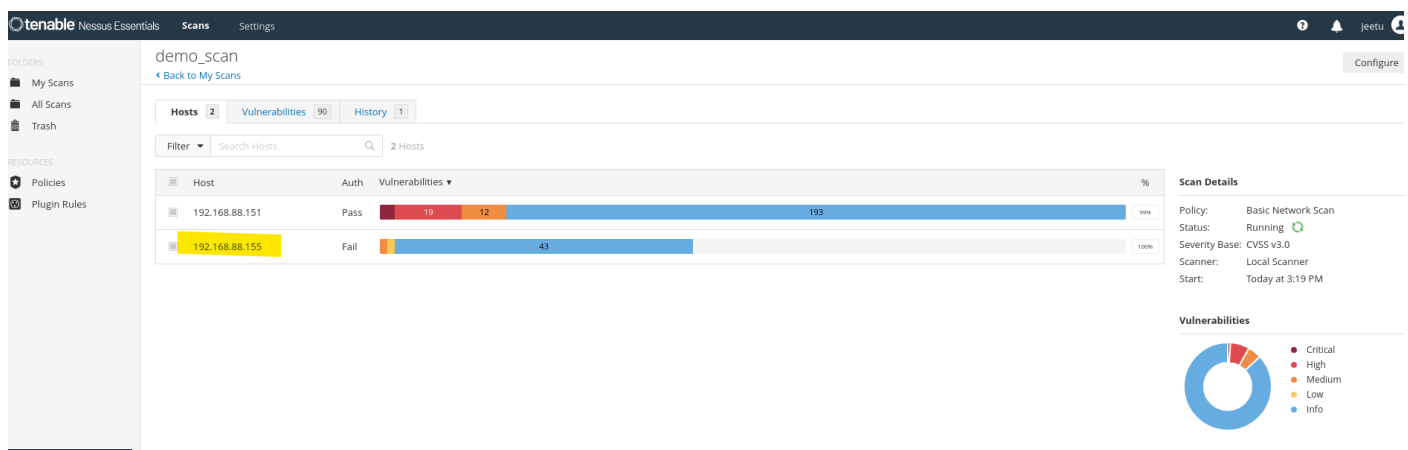
Goto Scan → New Scan



Choose Basic Network Scan



After scanning,



Once you click the target IP address, it will list all the vulnerabilities:

The screenshot shows the Tenable Nessus Essentials interface. The top navigation bar includes 'tenable', 'Nessus Essentials', 'Scans', and 'Settings'. The left sidebar has 'FOLDERS' (My Scans, All Scans, Trash) and 'RESOURCES' (Policies, Plugin Rules). The main content area is titled 'demo_scan / 192.168.88.155' with a 'Configure' button. Below the title is a 'Vulnerabilities' section showing 22 vulnerabilities. A table lists these vulnerabilities with columns for Severity, CVSS, VPR, EPSS, Name, Family, and Count. The table includes entries like 'SMB (Multiple Issues)', 'ICMP Timestamp Request Remote Date Disclosure', 'HTTP (Multiple Issues)', 'DCE Services Enumeration', 'Nessus SYN scanner', 'Service Detection', 'Common Platform Enumeration (CPE)', 'Device Type', 'Ethernet Card Manufacturer Detection', and 'Ethernet MAC Addresses'. A 'Host Details' sidebar on the right shows IP, MAC, OS, Start, End, Elapsed, and Auth information. A 'Vulnerabilities' donut chart is also present.

Sev	CVSS	VPR	EPSS	Name	Family	Count
MIXED	...	...	...	SMB (Multiple Issues)	Misc.	2
LOW	2.1 *	...	...	ICMP Timestamp Request Remote Date Disclosure	General	1
INFO	...	...	...	SMB (Multiple Issues)	Windows	6
INFO	...	...	...	HTTP (Multiple Issues)	Web Servers	5
INFO	...	...	...	DCE Services Enumeration	Windows	9
INFO	...	...	...	Nessus SYN scanner	Port scanners	5
INFO	...	...	...	Service Detection	Service detection	2
INFO	...	...	...	Common Platform Enumeration (CPE)	General	1
INFO	...	...	...	Device Type	General	1
INFO	...	...	...	Ethernet Card Manufacturer Detection	Misc.	1
INFO	...	...	...	Ethernet MAC Addresses	General	1

Click on vulnerabilities

The screenshot shows the Tenable Nessus Essentials interface. The top navigation bar includes 'tenable', 'Nessus Essentials', 'Scans', and 'Settings'. The left sidebar has 'FOLDERS' (My Scans, All Scans, Trash) and 'RESOURCES' (Policies, Plugin Rules). The main content area is titled 'demo_scan' with a 'Back to My Scans' link. Below the title is a 'Vulnerabilities' section showing 90 vulnerabilities. A table lists these vulnerabilities with columns for Severity, CVSS, VPR, EPSS, Name, Family, and Count. The table includes entries like 'CentOS (Multiple Issues)', 'Nagios Xi (Multiple Issues)', 'NFS Shares World Readable', 'Apache Tomcat (Multiple Issues)', 'SSL (Multiple Issues)', 'SMB (Multiple Issues)', 'ICMP Timestamp Request Remote Date Disclosure', 'HTTP (Multiple Issues)', 'SSH (Multiple Issues)', 'SMB (Multiple Issues)', and 'Mysql (Multiple Issues)'. A 'Scan Details' sidebar on the right shows Policy, Status, Severity Base, Scanner, and Start information. A 'Vulnerabilities' donut chart is also present.

Sev	CVSS	VPR	EPSS	Name	Family	Count
MIXED	...	...	...	CentOS (Multiple Issues)	CentOS Local Security Checks	22
MIXED	...	...	...	Nagios Xi (Multiple Issues)	CGI abuses	4
HIGH	7.5	...	...	NFS Shares World Readable	RPC	1
MIXED	...	...	...	Apache Tomcat (Multiple Issues)	Web Servers	8
MIXED	...	...	...	SSL (Multiple Issues)	General	10
MIXED	...	...	...	SMB (Multiple Issues)	Misc.	2
LOW	2.1 *	...	...	ICMP Timestamp Request Remote Date Disclosure	General	1
INFO	...	...	...	HTTP (Multiple Issues)	Web Servers	16
INFO	...	...	...	SSH (Multiple Issues)	General	8
INFO	...	...	...	SMB (Multiple Issues)	Windows	6
INFO	...	...	...	Mysql (Multiple Issues)	Databases	4

What Is Credentialed Scanning (Windows)?

Credentialed scanning allows Nessus to **log in to the Windows system** and perform **deep inspection**, including:

- Missing security patches
- Local vulnerabilities
- Weak configurations
- Installed software inventory
- Registry and policy checks

This provides **far more accurate results** than unauthenticated scans.

Settings → Credentials → Windows

Credential Options

Fill in:

- **Username:** Administrator or DOMAIN\user
- **Password:** Admin password
- **Authentication method:** Password
- **Elevate privileges:** Enabled

Optional:

- Enable **SMB Domain**
- Enable **Start Remote Registry service**

Step 3: Configure the Scan

1. Go to **Scans → New Scan**
2. Select **Basic Network Scan**
3. Enter target IP or hostname
4. Go to **Credentials tab**
5. Add **Windows credentials**
6. Save the scan

Step 8: What Credentialed Scanning Finds That Non-Credentialed Cannot

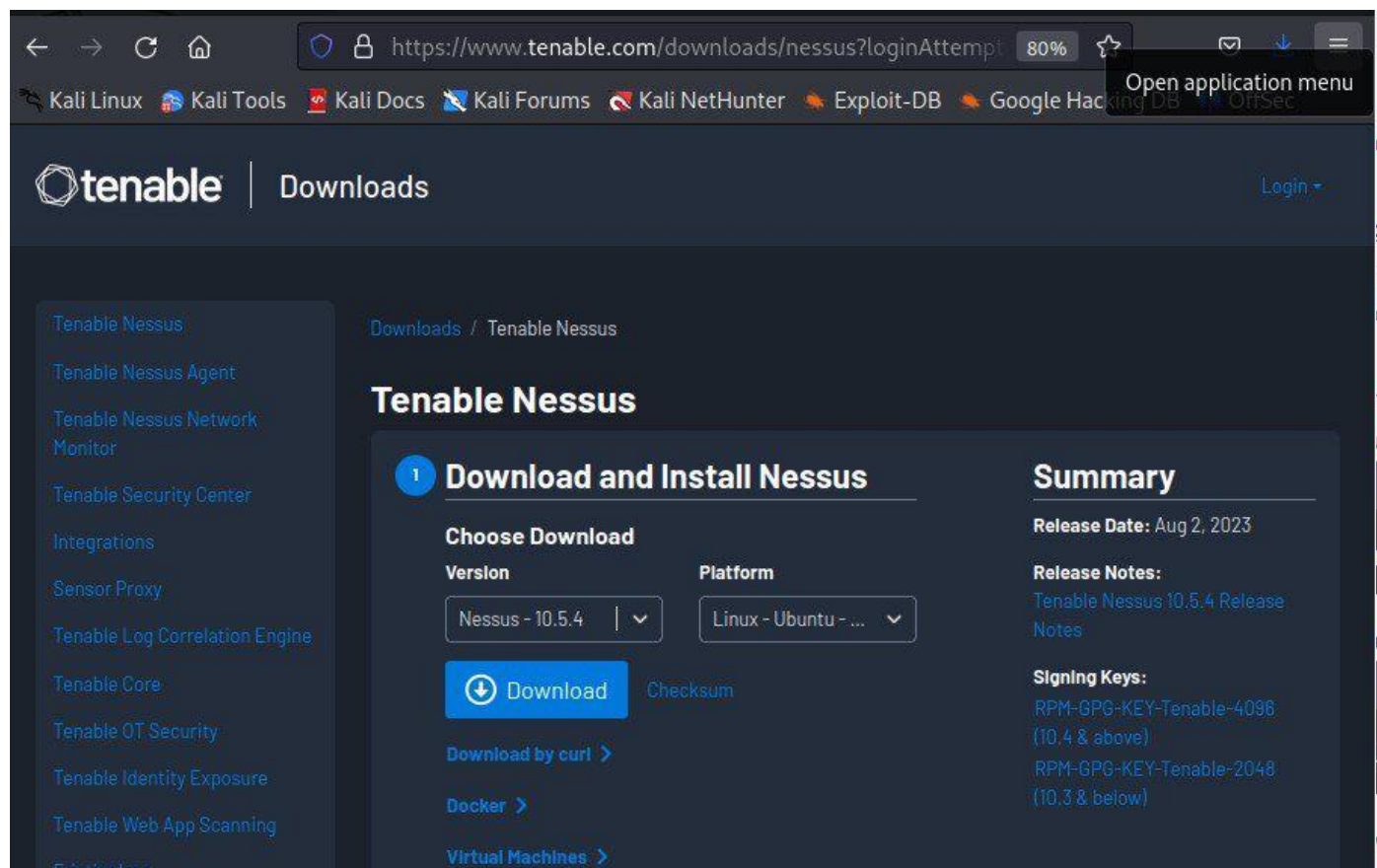
Area	Non-Cred	Credentialed
Missing patches	✗	✓
Local policies	✗	✓
Registry issues	✗	✓
Installed software	✗	✓
Accurate severity	✗	✓

Steps to Install Nessus on Kali

Step 1. Download Nessus:

Visit the [Tenable Nessus website](https://www.tenable.com/downloads/nessus) and sign up for a Nessus Home edition account (if you don't have one):

After signing up, you'll receive a link to download the Nessus package and also receive an activation code. Download the appropriate package for your Linux distribution.



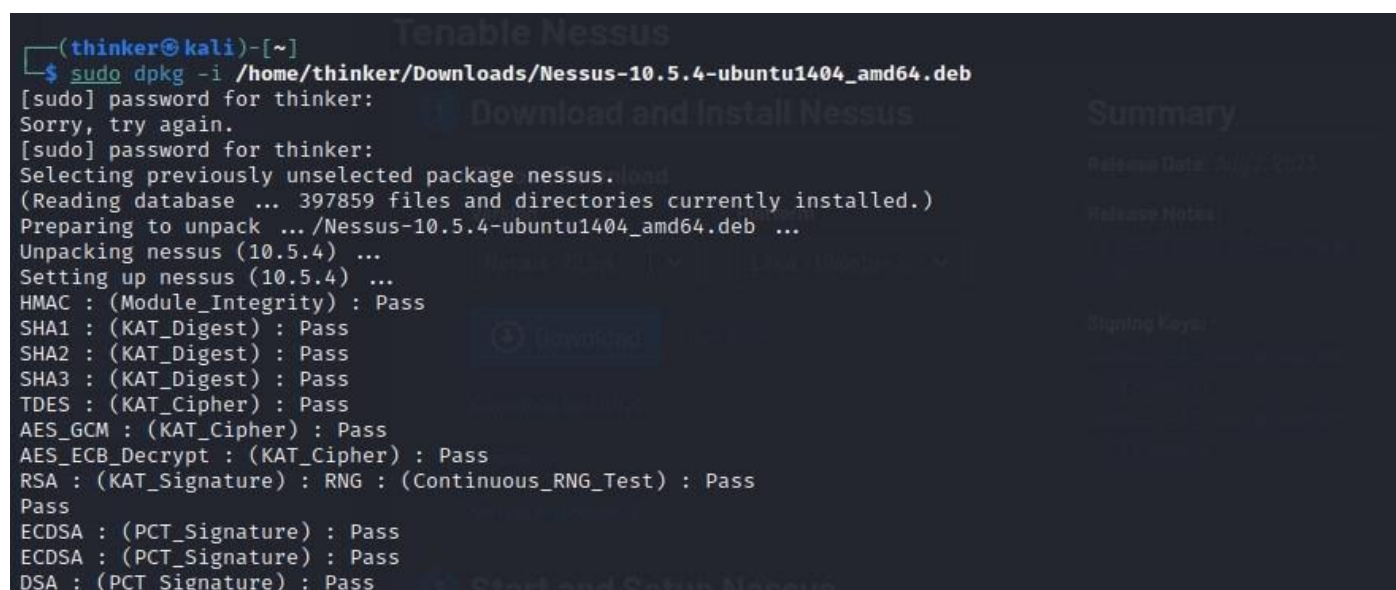
Download the appropriate package for your Linux distribution

Step 2. Install Nessus:

After downloading the Nessus package to your Downloads directory:

- **For Debian/Ubuntu:**

```
sudo dpkg -i /path/to/Nessus-<version>-<platform>.deb
```



Debian Installation

- **For CentOS/RedHat:**

```
sudo rpm -ivh /path/to/Nessus-<version>-<platform>.rpm
```

Note: Replace '**<version>**' with the version number you downloaded, and '**<platform>**' with the appropriate platform identifier.

Step 3. Start the Nessus Service:

After installation, start the Nessus service:

```
sudo systemctl start nessusd
```

Verify that the service is running by running the following command.

```
sudo systemctl status nessusd
```

```
(thinker@kali)-[~]
$ sudo systemctl status nessusd
● nessusd.service - The Nessus Vulnerability Scanner
   Loaded: loaded (/lib/systemd/system/nessusd.service; disabled; preset: disabled)
   Active: active (running) since Sat 2023-08-12 01:23:38 MDT; 15min ago
     Main PID: 18954 (nessus-service)
        Tasks: 15 (limit: 2226)
      Memory: 113.8M
         CPU: 1min 25.655s
       CGroup: /system.slice/nessusd.service
               └─18954 /opt/nessus/sbin/nessus-service -q
                 └─18956 nessusd -q
```

Running State-Disabled

If it is disabled then run the following command to enable it.

```
sudo systemctl enable nessusd
```

```
(thinker@kali)-[~]
$ sudo systemctl enable nessusd
Created symlink /etc/systemd/system/multi-user.target.wants/nessusd.service → /lib/systemd/system/nessusd.service.

(thinker@kali)-[~]
$
```

Now check whether it is enabled or not.

```
sudo systemctl status nessusd
```

```
(thinker@kali)-[~]
$ sudo systemctl status nessusd
● nessusd.service - The Nessus Vulnerability Scanner
   Loaded: loaded (/lib/systemd/system/nessusd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2023-08-12 01:23:38 MDT; 24min ago
     Main PID: 18954 (nessus-service)
        Tasks: 15 (limit: 2226)
      Memory: 91.8M
         CPU: 1min 32.776s
       CGroup: /system.slice/nessusd.service
               └─18954 /opt/nessus/sbin/nessus-service -q
                 └─18956 nessusd -q

Aug 12 01:23:38 kali systemd[1]: Started nessusd.service - The Nessus Vulnerability Scanner.
Aug 12 01:23:41 kali nessus-service[18956]: Cached 0 plugin libs in 0msec
Aug 12 01:23:41 kali nessus-service[18956]: Cached 0 plugin libs in 0msec

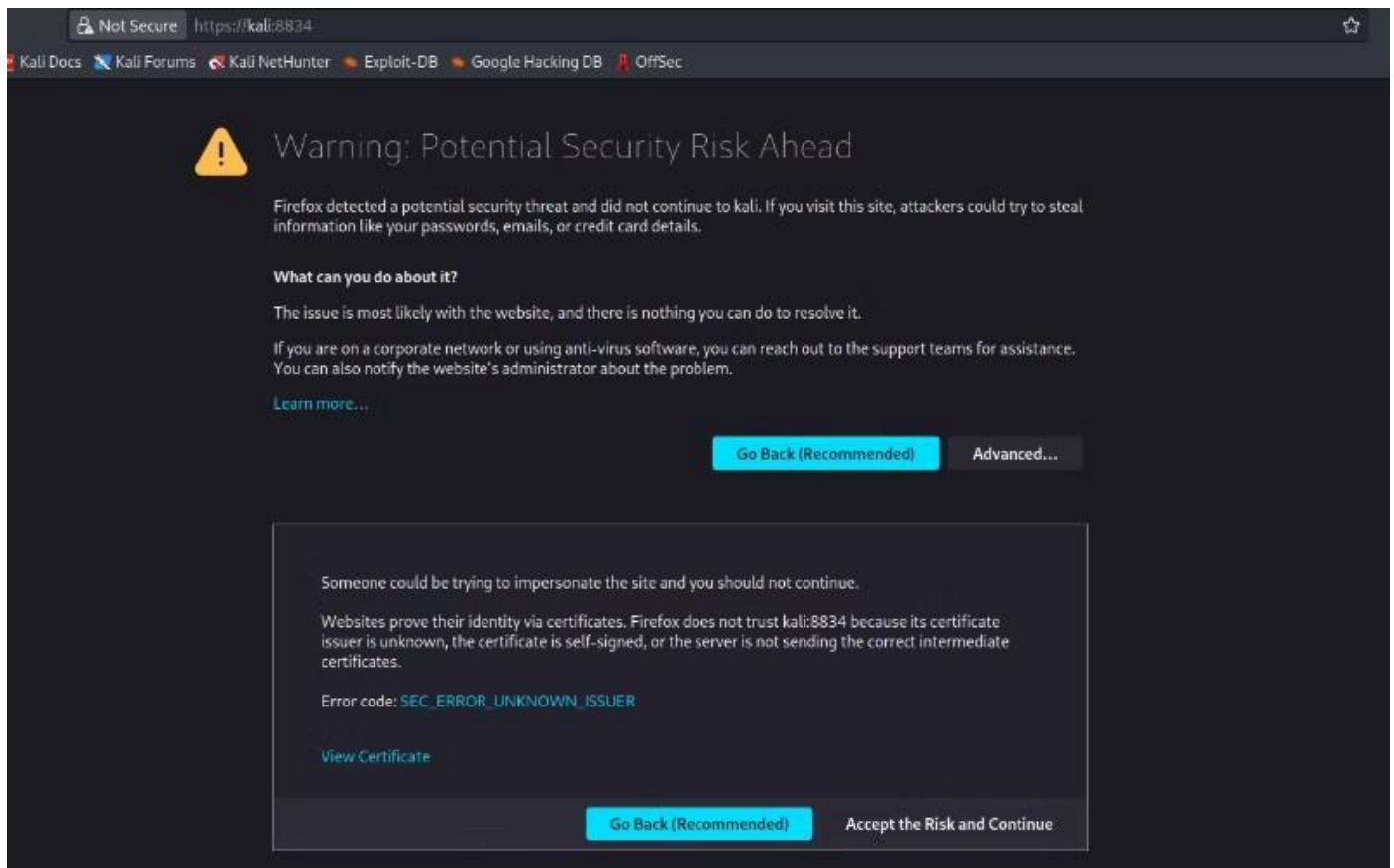
(thinker@kali)-[~]
$
```

Running State-Enabled

Step 4. Access the Nessus Web Interface:

Open a web browser and navigate to '<https://localhost:8834>'.

You might encounter a security warning due to the default self-signed SSL certificate used by Nessus. Accept the security risk and proceed.



Security Warning - accept the risk and proceed

Step 5. Activate Nessus:

On the Welcome to Nessus screen, select the product version, **Nessus Essentials**.



Welcome to Nessus

Choose how you want to deploy Nessus. Select a product to get started.

- ☒ Nessus Essentials
- ☐ Nessus Professional
- ☐ Nessus Manager
- ☐ Managed Scanner

Continue

Product Version

On the Get an activation code screen select skip, as you have previously obtained the activation code.



nessus[®]

Essentials

Get an activation code

To receive an email with a free Nessus Essentials activation code, enter your information.

If you already have an activation code, skip this step.

First *

John

Last *

Smith

Email *

user@example.com

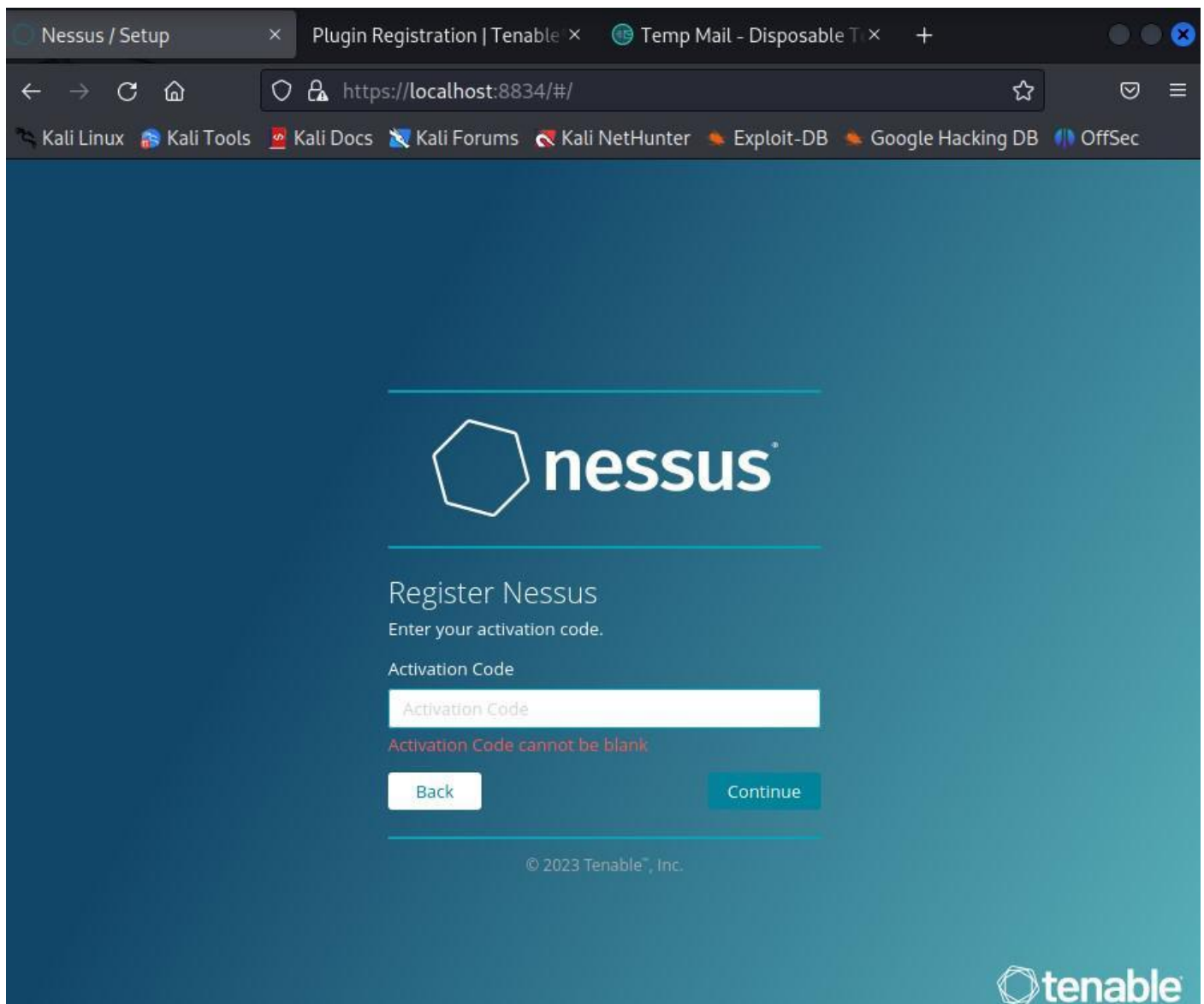
Skip

Back

Email

activation code

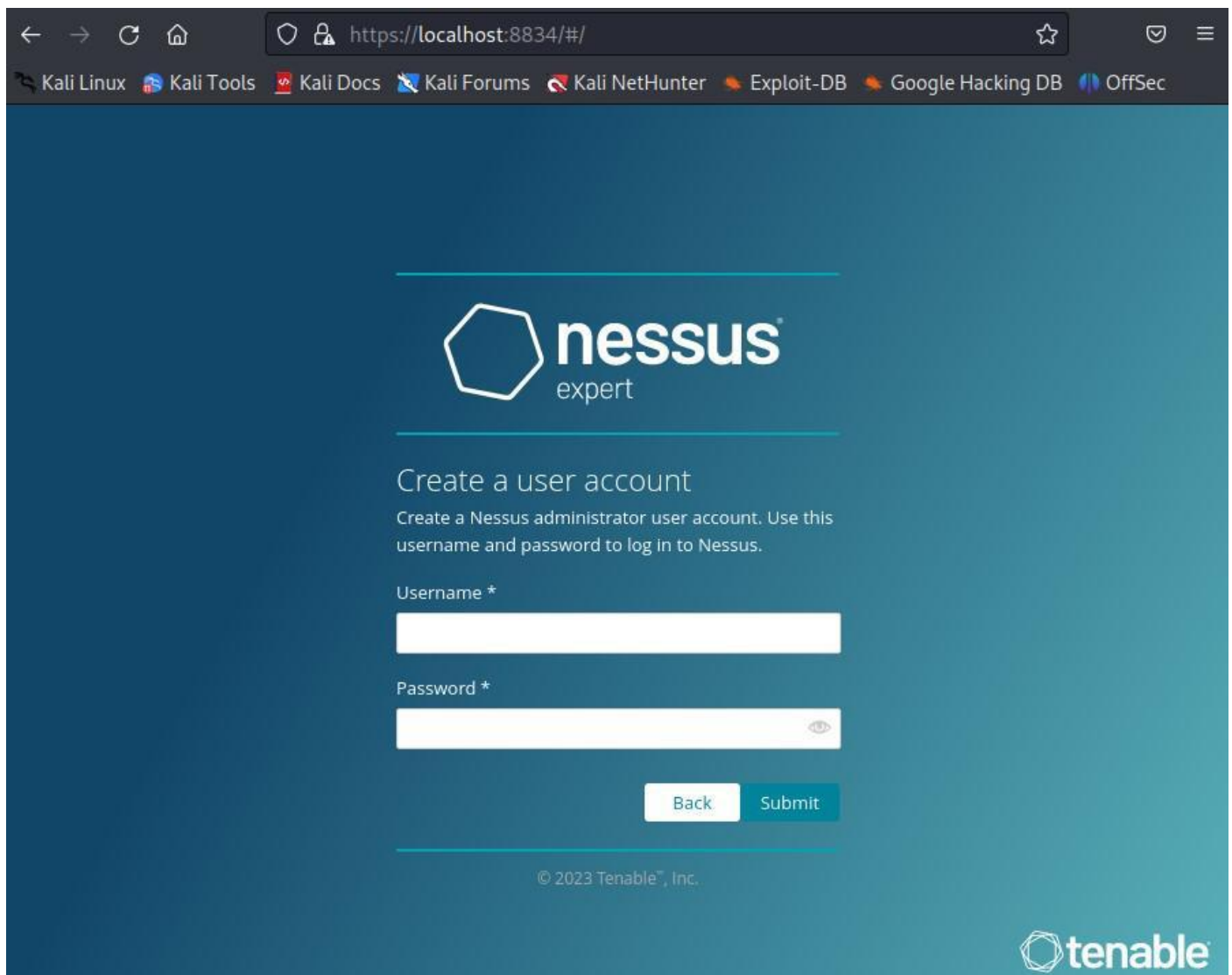
Enter the code to activate Nessus.



Enter the activation code

Step 6. Set Up a Username and Password:

After activation, Create a Nessus user account (**username and password**). This account will be used to access the Nessus web interface.



The screenshot shows a web browser window with the URL `https://localhost:8834/#/`. The browser's address bar and tabs are visible at the top. The main content area has a dark blue background with the Nessus Expert logo (a white hexagon) and the text "nessus expert". Below the logo, the heading "Create a user account" is displayed, followed by the instruction: "Create a Nessus administrator user account. Use this username and password to log in to Nessus." The form contains two input fields: "Username *" and "Password *". The "Password *" field has a toggle icon (an eye) to the right of the input box. At the bottom of the form are two buttons: "Back" and "Submit". The footer of the page includes the copyright notice "© 2023 Tenable™, Inc." and the Tenable logo.

← → ↻ 🏠 `https://localhost:8834/#/` ☆ 🛡️ ☰

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec

 **nessus**
expert

Create a user account

Create a Nessus administrator user account. Use this username and password to log in to Nessus.

Username *

Password *

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 **tenable**

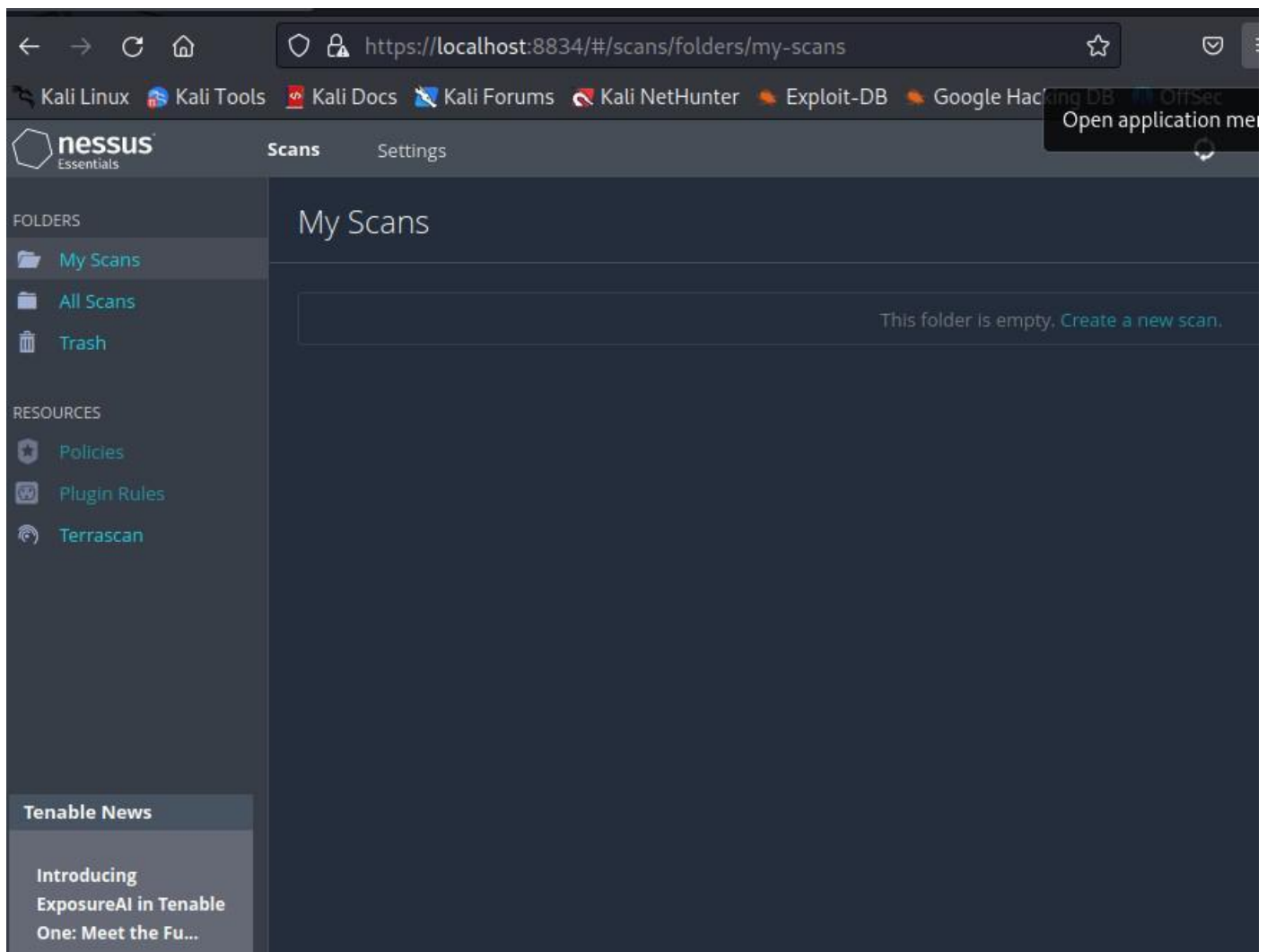
Setup Nessus User Account

Step 7. Log in to Nessus:

Log in to the Nessus web interface using the account you just created.

Step 8. Finish the Setup:

That's it! You now have Nessus installed and running on your Linux system. You can use the Nessus web interface to start scanning your network for vulnerabilities.



Nessus Web Interface

Let's go over two practical examples of using Nessus for vulnerability scanning.

How to Use Nessus on Linux?

Example 1: Basic scan

In this example, we will perform an initial vulnerability scan on the target system using Nessus.

Step 1: Start Nessus

Verify that you have turned on the Nessus service with the following command:

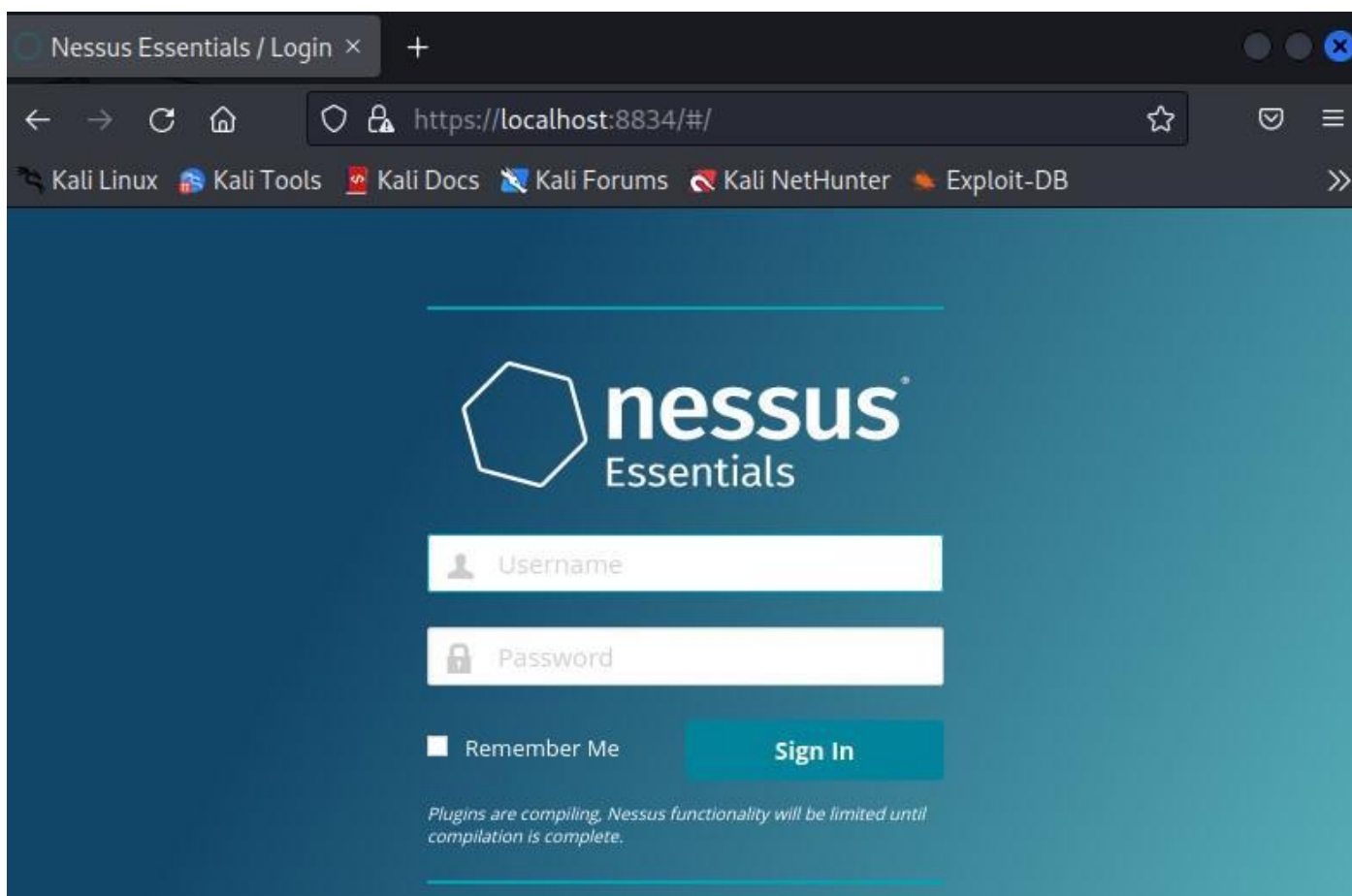
```
sudo systemctl start nessusd
```

```
thinker@kali: ~  
File Actions Edit View Help  
—(thinker@kali)-[~]  
$ ls  
Desktop Documents Downloads Music Pictures Public Templates Videos  
—(thinker@kali)-[~]  
$ sudo systemctl start nessusd  
[sudo] password for thinker:  
—(thinker@kali)-[~]  
$
```

Start Nessus

Step 2: Go to the Nessus Web Connection

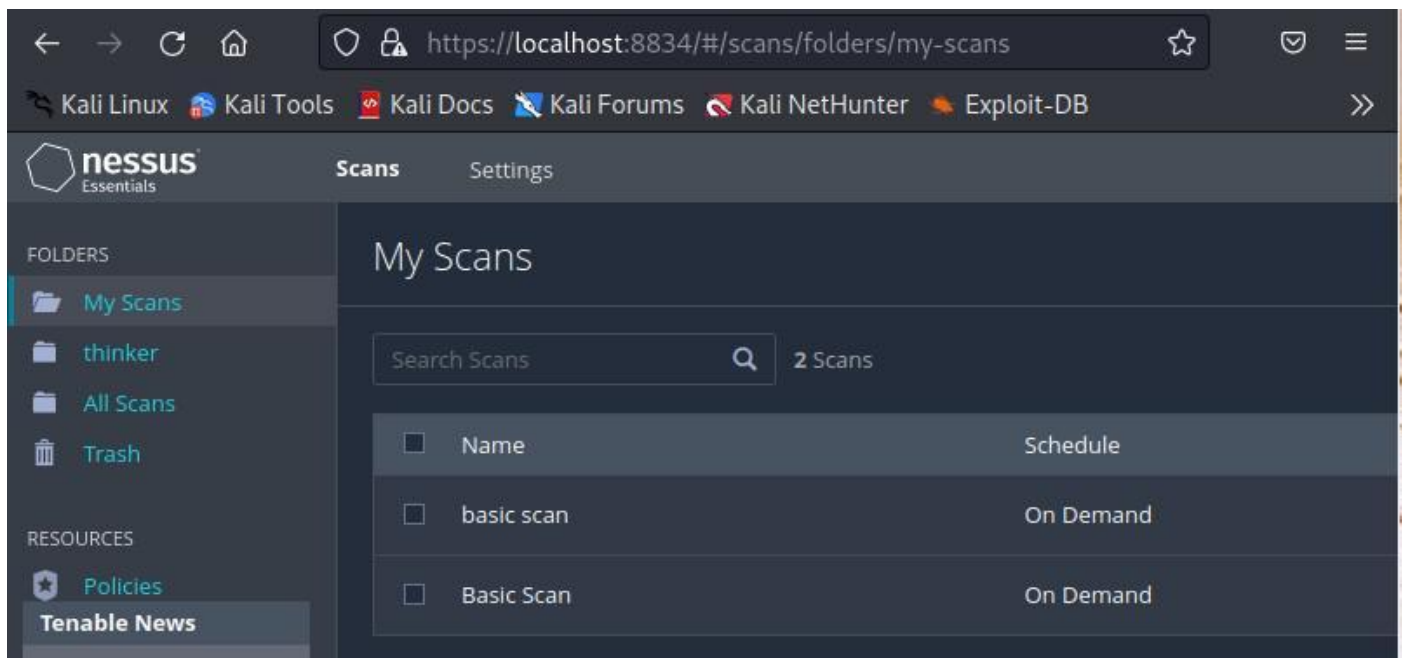
Open a web browser and navigate to **<https://localhost:8834>**. Log in with your Nessus username and password.



Log In Page

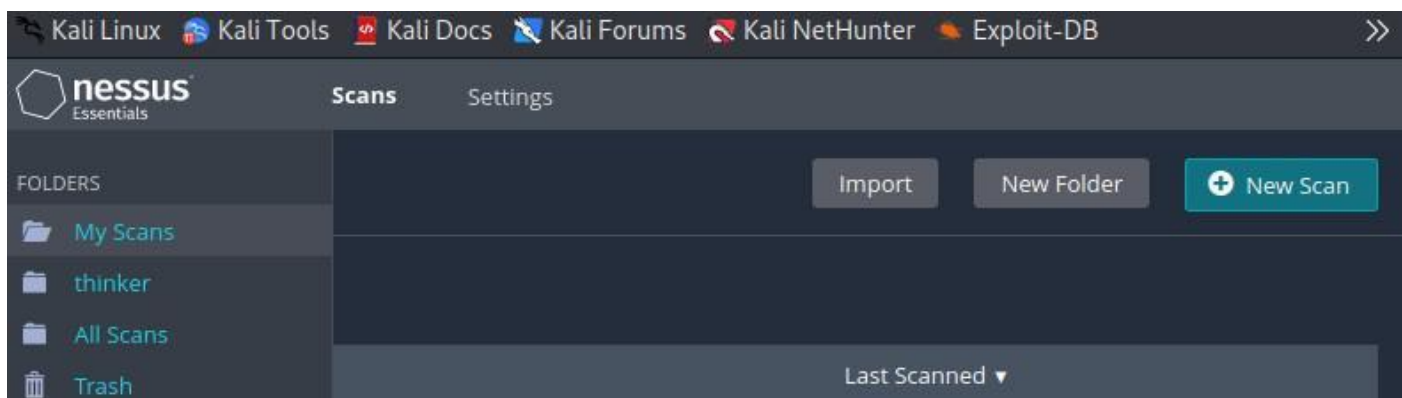
Step 3: Create a New Scan

1. Click on the "Scans" tab in the top navigation bar.



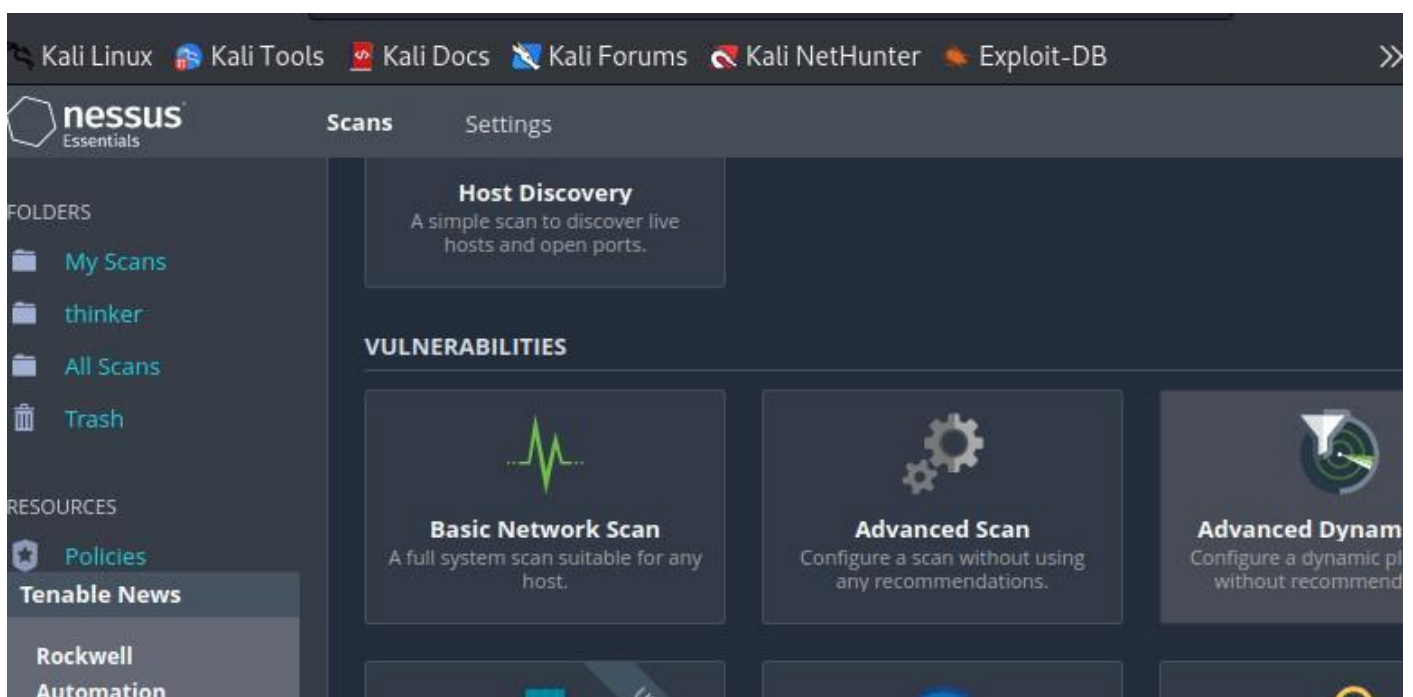
Scan Tab

2. Click the "New Scan" button.



Click on New Scan

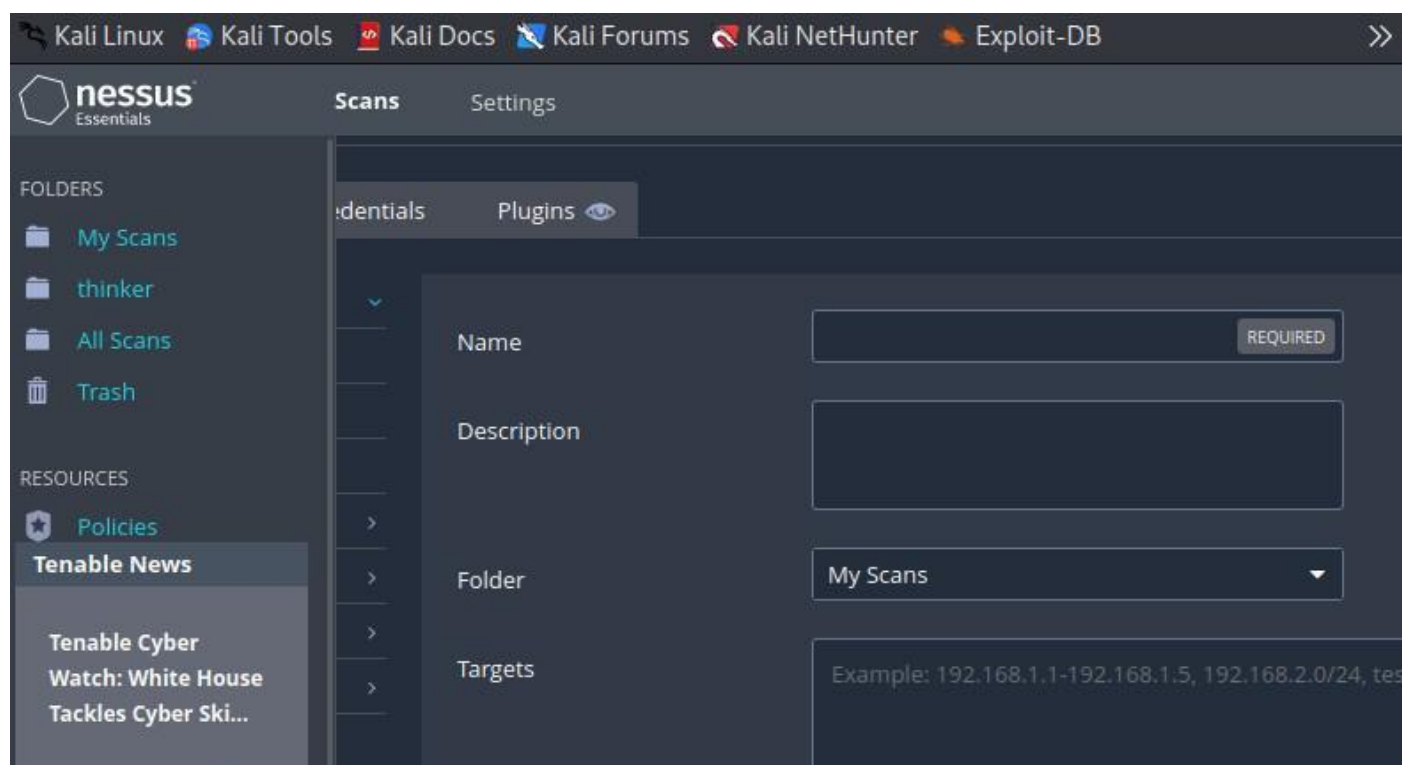
3. Select the "Basic Network Scan" template.



Templates

Step 4: Configure the Scan

In the **"Targets"** section, enter the [IP address](#) or hostname of the target system you want to scan. You can leave other settings at their default values for this basic example.



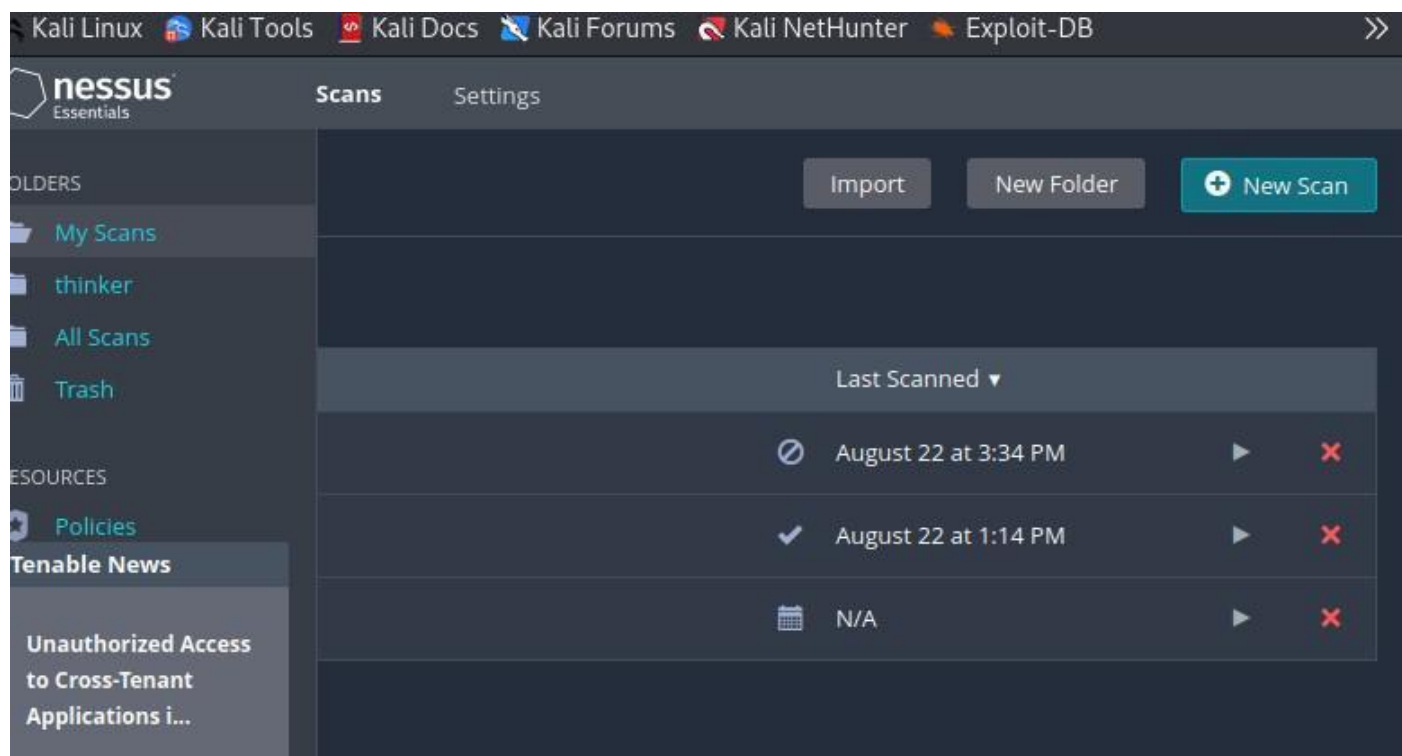
The screenshot shows the Nessus Scans configuration page. The 'Targets' field is highlighted with a red box. The 'Name' field is marked as 'REQUIRED'.

Field	Value
Name	
Description	
Folder	My Scans
Targets	Example: 192.168.1.1-192.168.1.5, 192.168.2.0/24, tes

Enter the scan name and target IP address

Step 5: Start the Scan

Click the **"Save"** button to save the scan configuration, then click the **"Launch"** button to start the scan.



The screenshot shows the Nessus Scans page with a list of scans. The 'Launch' button is highlighted with a red box.

Icon	Last Scanned	Play	Stop
⊘	August 22 at 3:34 PM	▶	✖
✓	August 22 at 1:14 PM	▶	✖
📅	N/A	▶	✖

Click on launch button

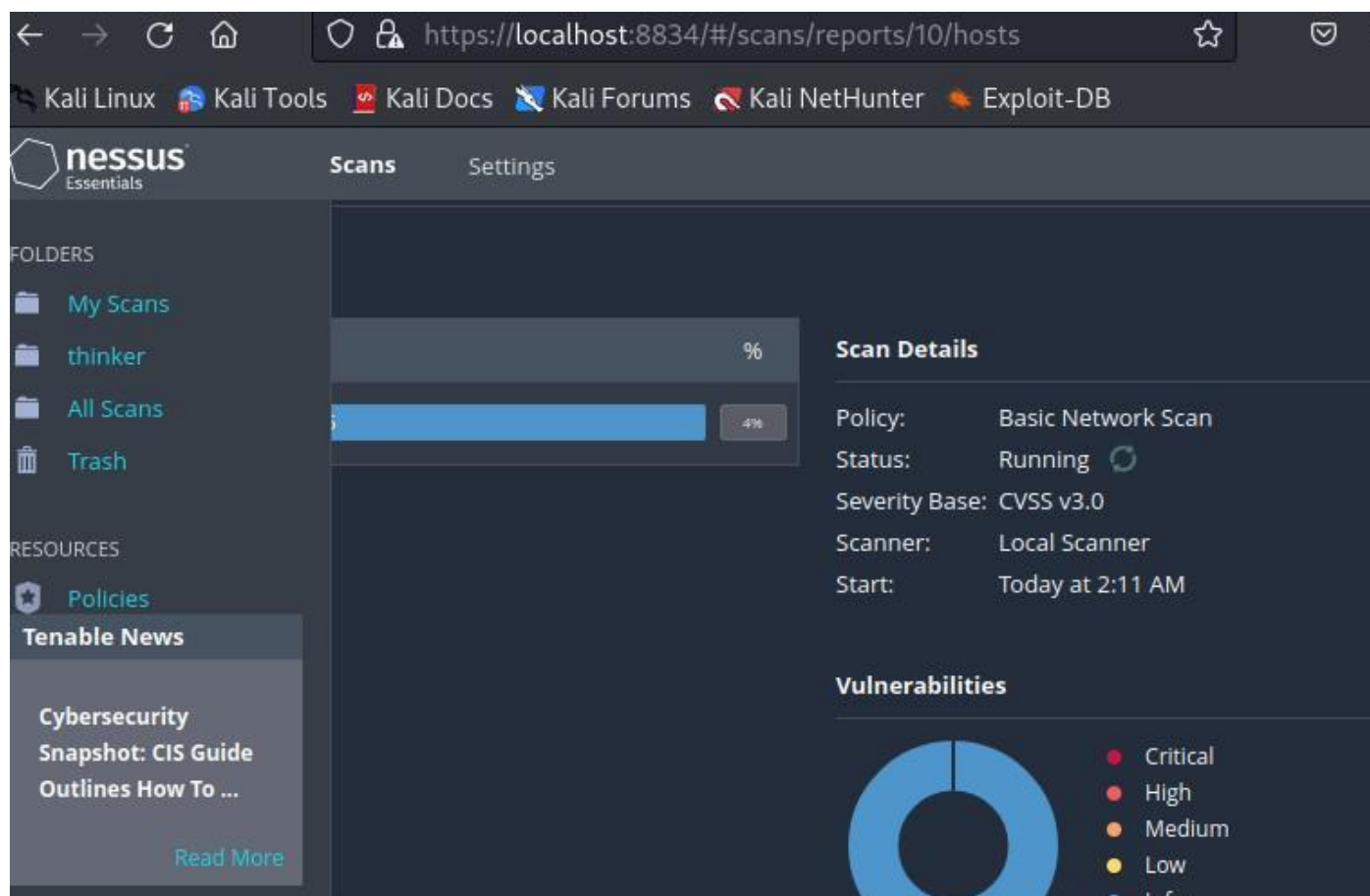
In this example, we started a basic network scan using Nessus. We provided the target's IP address or hostname and used the **"Basic Network Scan"** template. This scan looks for common vulnerabilities in the target system.

Example 2: Create a Report

In this example, we will create a report from the results of a previous scan.

Step 1: Wait for the scan to complete

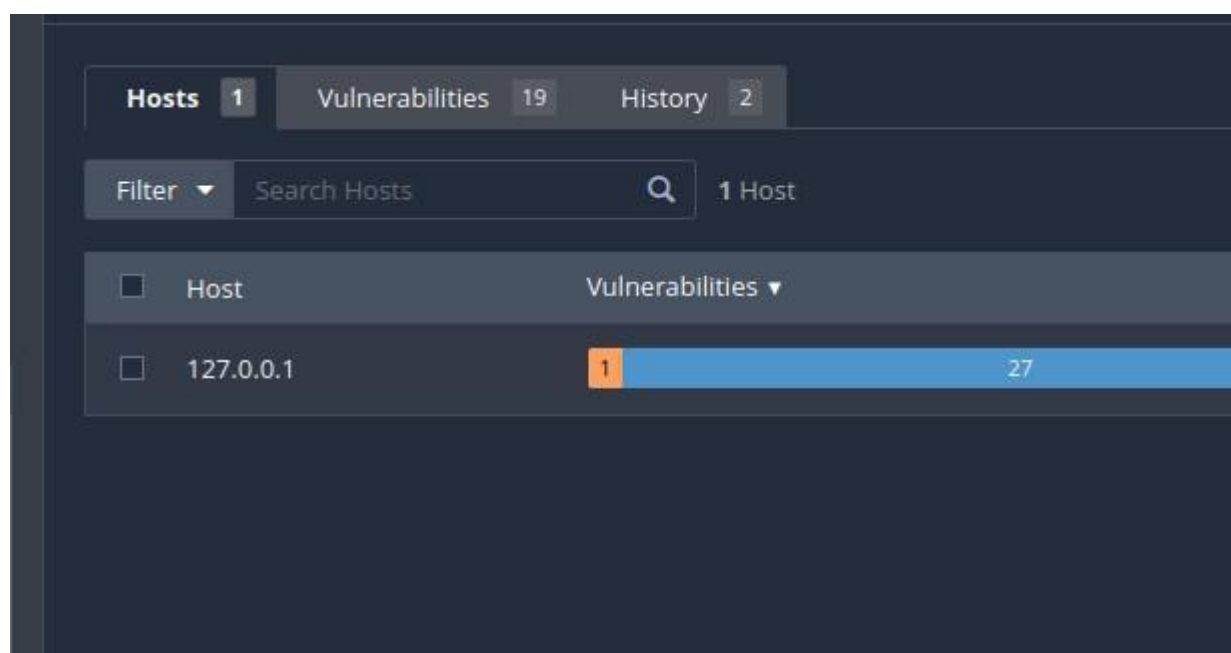
Wait for the scan to complete. Depending on the size of the target network, this can take some time.



Running Scan

Step 2: Access Scan Results

Once the scan is complete, go to the "Scans" tab. Click on the scan you performed earlier ("Basic Scan").



Scan tab

Step 3: Generate a Report

1. In the scan details page, click the "Export" button.

2. Select the format for the report (e.g., **PDF, HTML**).
3. Choose the report template (e.g., "**Executive Summary**").
4. Click the "**Generate**" button.

Generate Report

Report Format: ☐ HTML ☒ PDF ☐ CSV

Select a Report Template:

SYSTEM

Complete List of Vulnerabilities by Host

Detailed Vulnerabilities By Host

Detailed Vulnerabilities By Plugin

Vulnerability Operations

Template Description:

This report provides a summary list of vulnerabilities detected in the scan.

Filters Applied:

None

Formatting Options:

☒ Include page breaks between vulnerability results

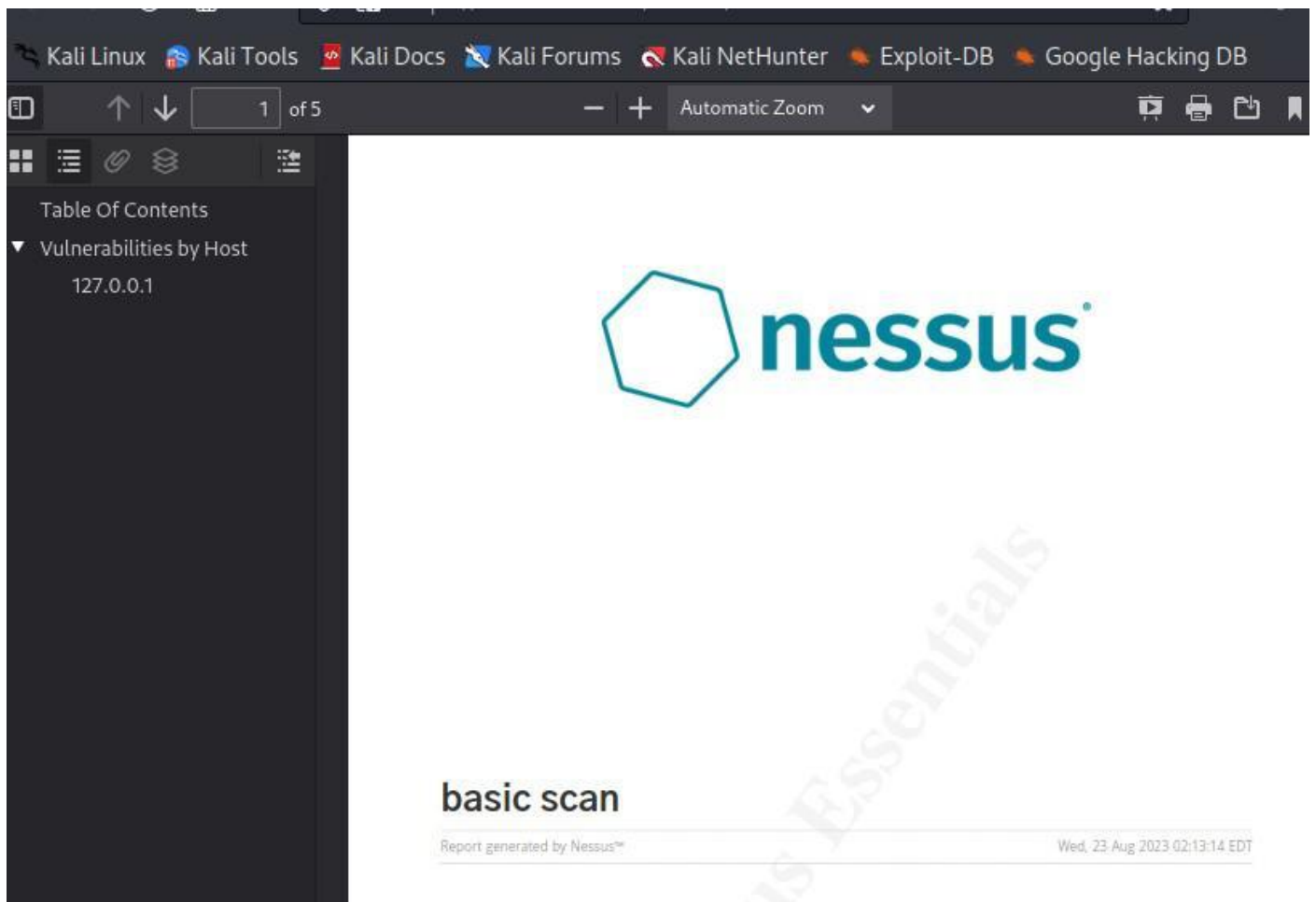
Generate Report

Cancel

Generate a Report

Step 4: Download the Report

Once the report is generated, you'll see a link to download it. Click the link to download the report in the selected format.



In this example, we get the scan results and generate a report. Nessus offers a variety of report templates, such as **"Executive Summary"** and **"Full Audit"**.